Regional/Global Warming and Statistics: A statisticians view of the surge in warming debate

Rebecca Killick(killick@clemson.edu)
Joint work with Claudie Beaulieu (Uni California Santa Cruz)
Parts with: Adelicia Johnson, Colin Gallagher, Robert Lund, Xueheng Shi
ICSA Montreal Aug 2026

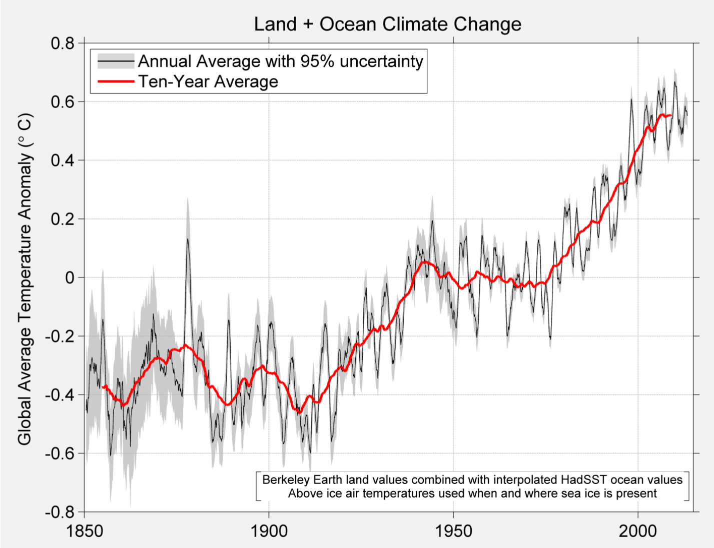
Come work with me!

- Jeff and Karen Camm Endowed Professorship in Data Science and Decision Intelligence
- Tenure Track Assistant Professor level
- Start Aug 15 2027
- Submit by 15th Oct 2026
- Open to traditional and non-traditional backgrounds

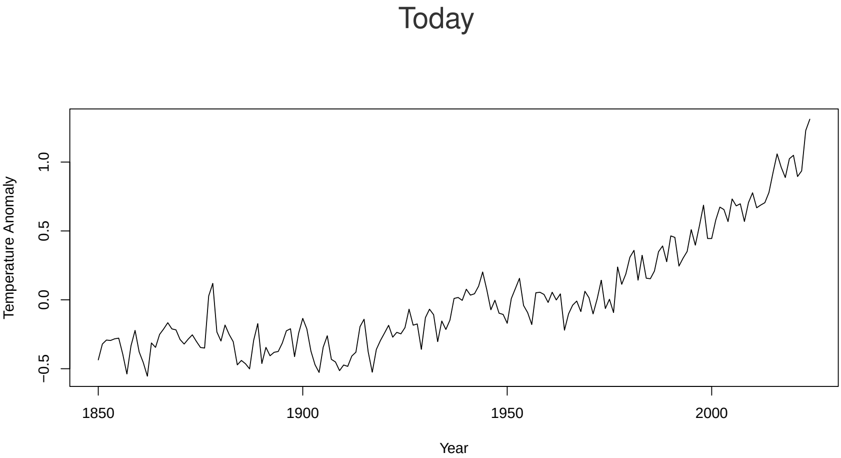
Outline

- What is the warming debate in climate science?
- What are changepoints?
- The story of a statistical solution
- Some frustrations

Global Mean Temp 2016



Global Mean Temp



Where is Statistics?

CLEMSON UNIVERSITY

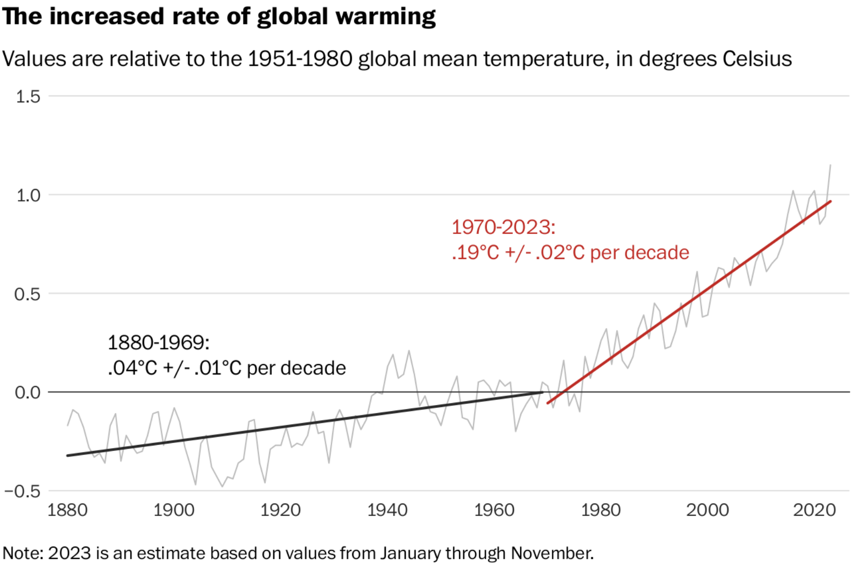
Is climate change speeding up? Here's what the science says.

This year's record temperatures have some scientists concerned that the pace of warming may be accelerating. But not everyone agrees.

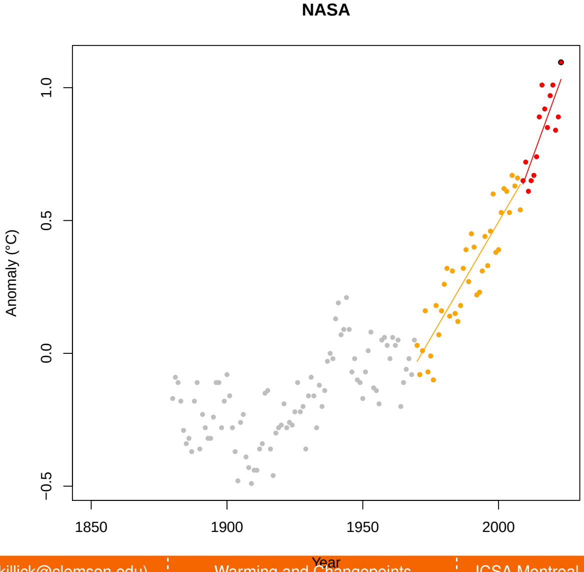
By Chris Mooney and Shannon Osaka
December 26, 2023 at 6:30 a.m. EST

For the past several years, a small group of scientists has warned that sometime early this century, the rate of global warming — which has remained largely steady for decades — might accelerate. Temperatures could rise higher, faster. The drumbeat of weather disasters may become more insistent.

Where is Statistics?



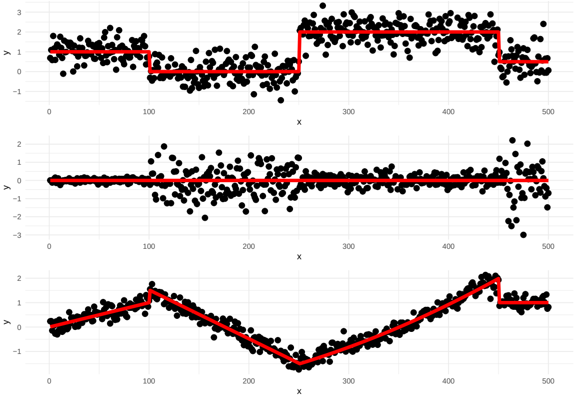
Where is Statistics? NYT



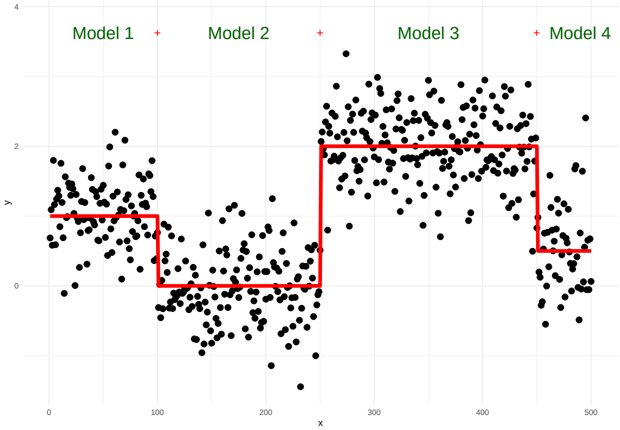
What are changepoints?

For data $y_1, \ldots, y_n$, if a changepoint exists at τ, then $y_1, \ldots, y_\tau$ differ from $y_{\tau+1}, \ldots, y_n$ in some way.

There are many different types of change.



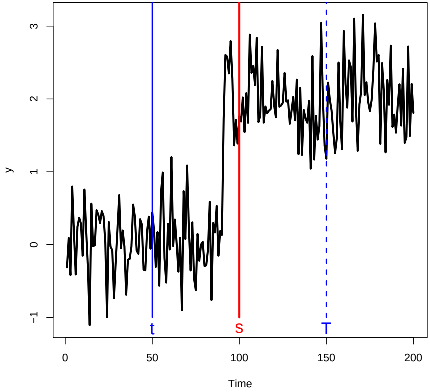
Problem



- How many changes?
- Where are the changes? 2^{n-1} possible solutions!

PELT in a nutshell

- Dynamic programming allows us to only worry about the location of the *last* change.
- Pruning means that as we go through the data we are smart about which locations are potential last change locations.

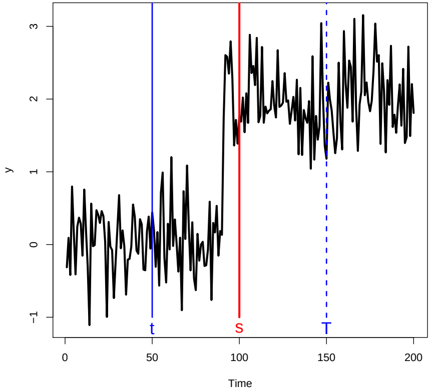


PELT: Pruning

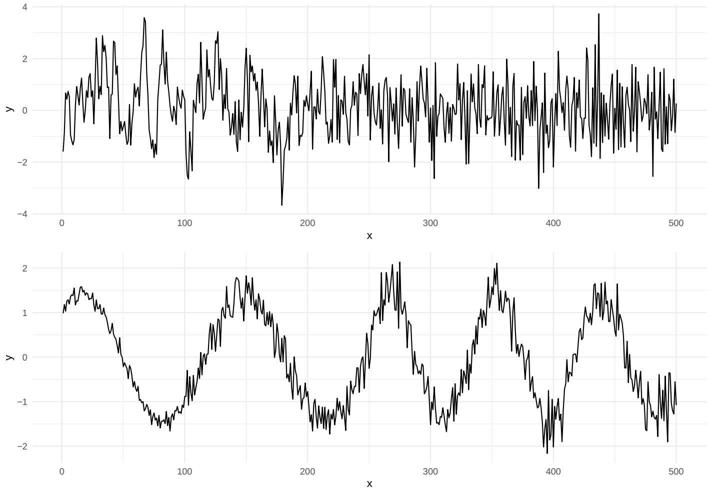
Let $0 < t < s < T$, if

$$F(t) + \mathcal{C}(y_{(t+1):s}) < F(s)$$

then at any future time $T > s$, t can never be the optimal last changepoint prior to T.
We can prove that, under certain regularity conditions, the expected computational complexity will be $\mathcal{O}(n)$.

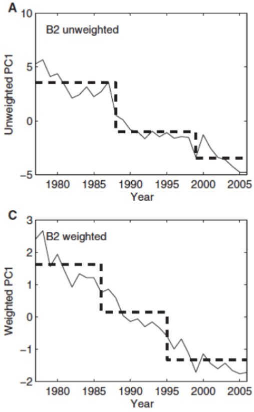


More complicated changes



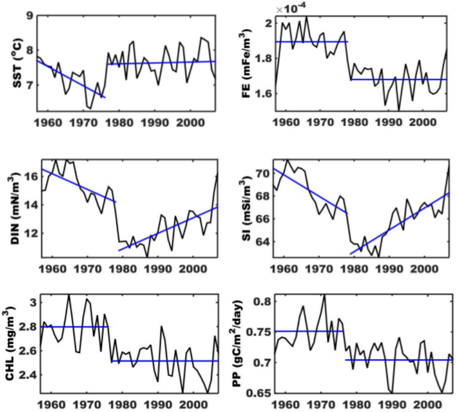
Tooling not enough

- From a publication in Marine Ecology (not the only one)
- Used the Rodionov (2004) method.
- Very popular but cannot deal with trend or autocorrelation.



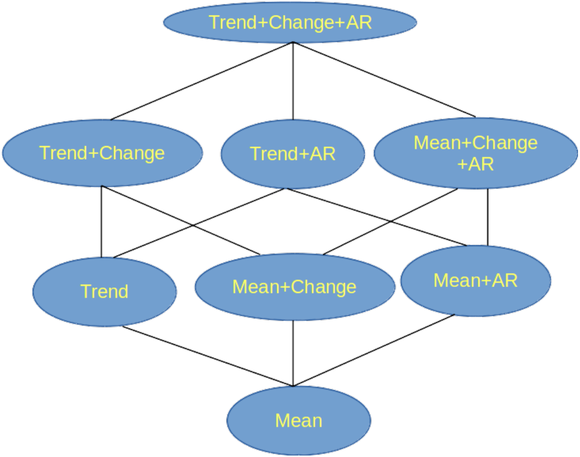
Beaulieu et al. 2015

- potentially hundreds or thousands of series
- no time to consider the format of change for each
- need to include both the potential for trends and also red noise (autocorrelation).

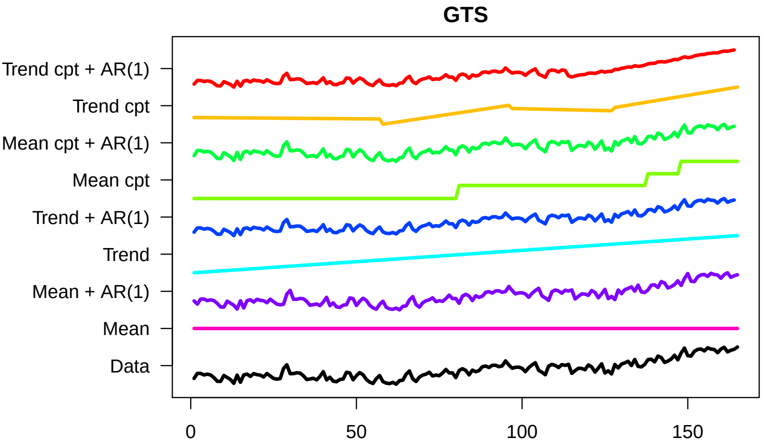


Model Selection

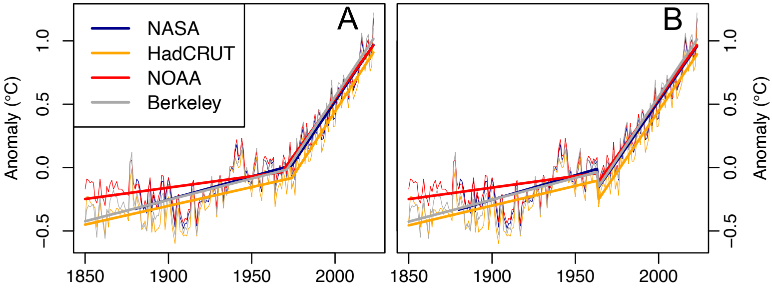
`EnvCpt`: select the most parsimonious but accurate model for the data. Simple to extend with other types of models.



GMST All Fits



Join-pin Models



Join-Pin IID

Fearnhead, Maidstone, Letchford, JCGS (2019)

$$Y_t = \theta_i + \frac{\theta_{i+1} - \theta_i}{\tau_{i+1} - \tau_i}(t - \tau_i) + Z_t$$

where $Z_t \sim \mathrm{N}(0, \sigma^2)$.

Implemented in the `cpop` R package.

Join-Pin AR(p)

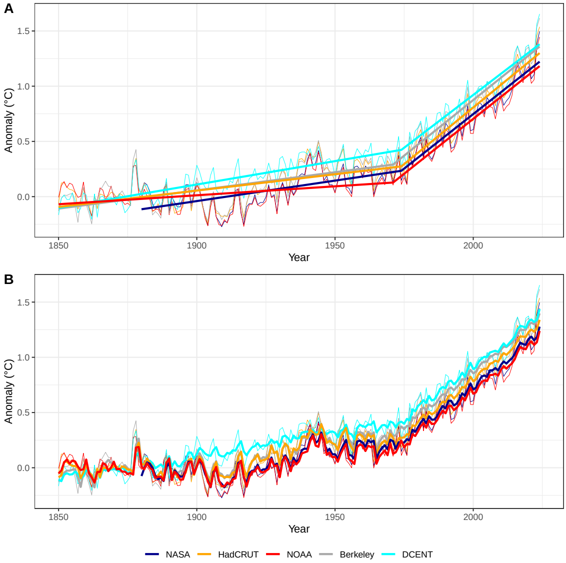
$$Y_t = \theta_i + \frac{\theta_{i+1} - \theta_i}{\tau_{i+1} - \tau_i}(t - \tau_i) + Z_t$$

but now Z_t is AR(p), $Z_t = \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + \ldots + \phi_p Z_{t-p} + \epsilon_t$.

Challenge:

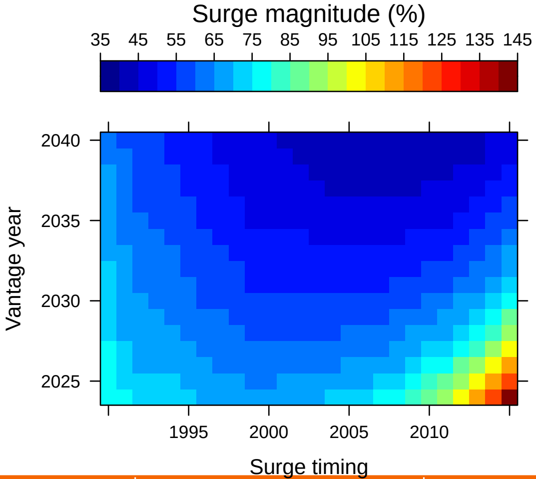
- Definition of join-pin
- Fitting AR parameter across segments:
  - Fixed we use EM algorithm (ask me about new method presented at JSM)
  - Varying we can embed in PELT

Join-pin Models



So what?

What would we need to see (in the statistically preferred model).



The fallout

Global warming is NOT surging, scientists say - despite record-breaking temperatures

- **Scientists find no evidence for a 'surge' in global temperatures**
- **READ MORE: 2023 was officially the hottest year on RECORD**

By JONATHAN CHADWICK FOR MAILONLINE
PUBLISHED: 14:01, 15 October 2024 | **UPDATED:** 07:52, 16 October 2024

Share 138 shares 407 View comments

From the **UK's hottest day** to the **hottest year on record globally**, there's no doubt some worrying temperature records have been broken in recent years.

Many people think the rate of **global warming** has dramatically accelerated or 'surged' over the past 15 years – and is a cause of more extreme weather.

But a new study says there is not any statistical evidence for this so-called 'surge' or 'leap'.

Stoney-McBoney · 2mo ago ·

This is that article your uncle is going to bring up next Thanksgiving as to why "all this global warming stuff is liberal nonsense".

30 Reply Award Share ···

Siphilius · 2mo ago ·

Well, if there's no discernible proof that we've attributed to an increase in climate change, as this paper suggests, he would be correct wouldn't he?

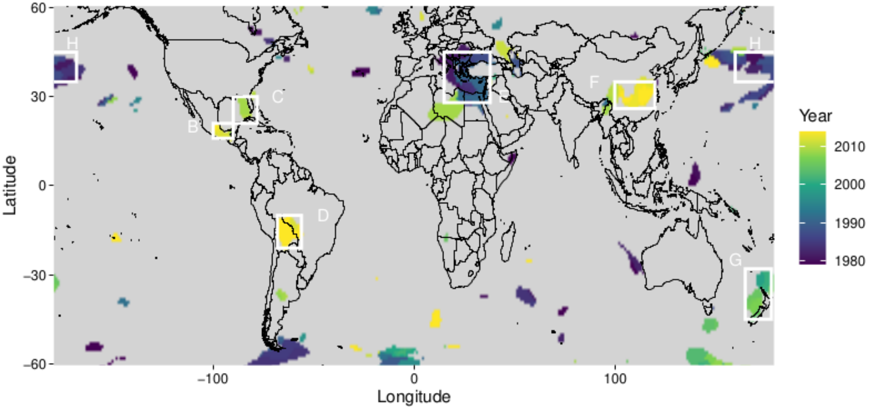
-36 Reply Award Share ···

Aacron · 2mo ago ·

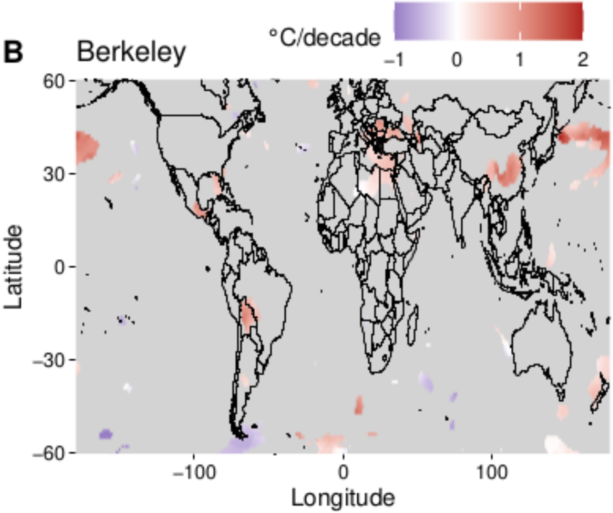
That's not what the article says, so no he wouldn't be.

24 Reply Award Share ···

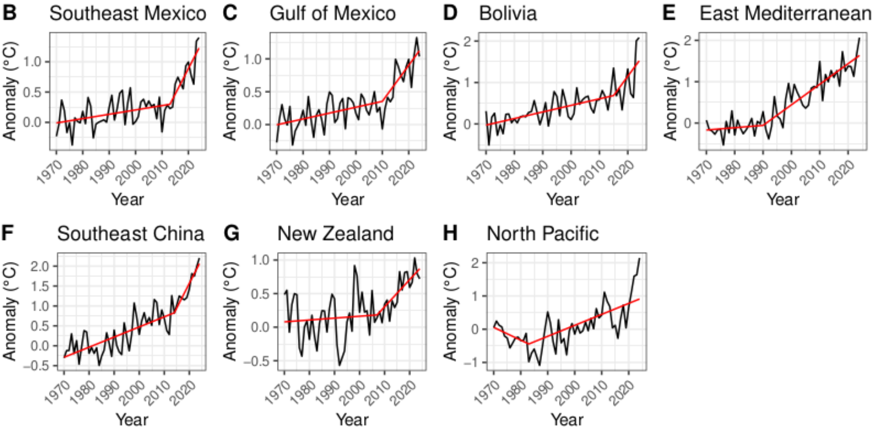
Extension to gridded



Extension to gridded



Extension to gridded



Summary

CLEMSON UNIVERSITY

- Detecting and documenting changepoints improves analyses . . .
- . . . and can help answer pertinent questions in different domains
- Extending to more complex model structures is interesting statistically
- Extending to monthly/grid point is interesting climatically

- I enjoy working with different disciplines . . .
- . . . as often it sparks my next research challenge.

Papers discussed

Preprints of all available at: www.lancs.ac.uk/~killick/pub.html

PELT: https://doi.org/10.1080/01621459.2012.737745

Model Choice: https://doi.org/10.1175/JCLI-D-17-0863.1 & https://doi.org/10.1002/qre.2712

LMvsCpts: https://doi.org/10.1007/s11222-017-9731-0 & https://doi.org/10.1002/env.2568

Warming Surge: https://arxiv.org/abs/2403.03388

Regional Surge: https://www.researchsquare.com/article/rs-7731926/v1

Climate Dynamics